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 CBSE Class 12 Chemistry BY @cbsesirji

100 Important 5-Mark Questions

(Based on Previous Year Board Trends)

♦ PHYSICAL CHEMISTRY

1. Explain how a buffer solution resists changes in pH and derive the Henderson–Hasselbalch equation.
2. Discuss Raoult's law and its application to ideal solutions.
3. Calculate lowering of vapour pressure and elevation of boiling point (numerical).
4. Define osmotic pressure and derive its expression.
5. Differentiate between molarity and molality and discuss temperature effects.
6. Derive the Nernst equation and explain its applications.
7. Describe construction and working of lead–acid battery.
8. Explain measurement of conductivity and effect of dilution.
9. Describe fuel cells and compare them with conventional cells.
10. Explain electrolysis of molten NaCl and aqueous CuSO₄.
11. Derive integrated rate law for first-order reaction and half-life expression.
12. Explain Arrhenius equation and temperature effect on rate.
13. Differentiate order and molecularity with examples.
14. Define activation energy and explain catalytic action.
15. Derive rate law and half-life for second-order reaction.
16. Explain adsorption isotherms and factors affecting adsorption.
17. Compare homogeneous and heterogeneous catalysis.
18. Describe formation and properties of colloids.
19. Differentiate physical and chemical adsorption.
20. Explain applications of surface chemistry in water purification.

♦ INORGANIC CHEMISTRY

21. Describe extraction of iron in blast furnace with reactions.
22. Explain Mond process for refining nickel.
23. Describe Bayer's process for aluminium extraction.
24. Explain zone refining and its applications.
25. Describe separation of noble gases from air.
26. Discuss trends in properties of group 15 elements.

27. Explain inert pair effect with examples.
28. Write notes on interhalogen compounds.
29. Discuss structure and acidity of oxoacids of sulphur.
30. Explain disproportionation reactions with examples.
31. Define lanthanide contraction and its consequences.
32. Compare lanthanoids and actinoids.
33. Explain formation and properties of coordination compounds.
34. Discuss colour and magnetic properties of transition metal complexes.
35. Explain chelation and stability of complexes.
36. Discuss isomerism in coordination compounds.
37. Explain Werner's theory of coordination compounds.
38. Describe applications of coordination compounds.
39. Discuss bonding in $[\text{Fe}(\text{CN})_6]^{4-}$ complex.
40. Explain chelate effect with examples.

♦ ORGANIC CHEMISTRY

41. Explain nucleophilic substitution mechanisms in haloalkanes.
42. Describe preparation and reactions of alkyl halides.
43. Explain effect of leaving group on substitution reactions.
44. Discuss environmental concerns related to haloarenes.
45. Predict products of haloalkane conversion reactions.
46. Explain mechanism of dehydration of alcohols.
47. Discuss acidic nature of phenols with resonance.
48. Write reactions of ethers.
49. Describe preparation and structure of ethers.
50. Explain uses and reactions of phenol.
51. Explain nucleophilic addition in aldehydes and ketones.
52. Discuss oxidation reactions of aldehydes and ketones.
53. Describe Hell–Volhard–Zelinsky reaction with mechanism.
54. Explain preparation and reactions of carboxylic acids.
55. Solve organic interconversion problems.
56. Explain basicity of amines.
57. Describe Gabriel phthalimide synthesis and limitations.
58. Explain electrophilic substitution in aromatic amines.
59. Convert aniline to p-bromoaniline.
60. Arrange amines in order of basic strength with reasons.
61. Write notes on carbohydrates and classify them.
62. Explain structure and functions of proteins.
63. Discuss properties and significance of enzymes.
64. Describe vitamins and deficiency diseases.
65. Differentiate DNA and RNA structures.

66. Classify polymers with examples and uses.
67. Explain addition and condensation polymerisation.
68. Describe biodegradable polymers.
69. Explain coordination polymers with examples.
70. Discuss environmental impact of plastics.
71. Differentiate antiseptics and disinfectants.
72. Explain role of chemicals in food preservation.
73. Describe drug action and side effects of analgesics.
74. Explain chemistry of detergents.
75. Describe role of chemistry in health and hygiene.
76. Convert ethanol to ethanoic acid with reagents.
77. Convert benzene to nitrobenzene and aniline.
78. Identify functional group and outline reactions of given compound.
79. Predict product and mechanism of electrophilic addition.
80. Explain retrosynthesis of a complex organic molecule.

♦ BOARD-IMPORTANT / VALUE-BASED

81. Discuss principles of green chemistry.
82. Explain reverse osmosis in water purification.
83. Describe role of chemistry in pollution control.
84. Explain corrosion and its prevention.
85. Discuss hardness of water and softening methods.
86. Write note on safe disposal of laboratory chemicals.
87. Describe municipal water treatment methods.
88. Discuss environmental impact of metal extraction industries.
89. Explain industrial role of catalysts.
90. Discuss sustainable use and recycling of polymers.
91. Explain chromatography principles and applications.
92. Describe ligand substitution reactions.
93. Discuss activation energy in real-life reactions.
94. Explain industrial applications of chemical equilibrium.
95. Discuss role of organic chemistry in drug discovery.
96. Explain role of electrolytes in biological systems.
97. Describe buffer action in blood.
98. Discuss factors affecting reaction rate in industry.
99. Explain medicinal importance of coordination compounds.
100. Describe importance of chemistry in household products.